arnavv@mit.edu

Arnav Verma

Education

Massachusetts Institute of Technology *Cambridge, United States*

Doctor of Philosophy in Computer Science and Electrical Engineering *Sept. 2025 - current*

- Thesis Area: Visualization and Cognitive Science.
- Thesis Advisor: Arvind Satyanarayan

University of Toronto *Toronto, Canada*

Master of Science in Computer Science *Sept. 2021 - April 2023*

- Thesis Area: Visualization and Psychology.
- Thesis Advisor: Fanny Chevalier

University of Toronto *Toronto, Canada*

Honors Bachelor of Science in Computer Science *Sept. 2016 - April 2021*

- Minor in Statistics
- Minor in Mathematics

Manuscripts and Publications

Verma, A., Mukherjee, K., Potts, C., Kreiss, E., and Fan, J. (2026). Human-centered evaluation of data visualization understanding in vision-language models. *Scientific Reports*.

Verma, A. and Fan, J. E. (2025). Measuring and predicting variation in the difficulty of questions about data visualizations. *Proceedings of the 47th Annual Meeting of the Cognitive Science Society*.

Brockbank, E., **Verma, A.**, Lloyd, H., Huey, H., Padilla, L., and Fan, J. E. (2025). Evaluating convergence between two data visualization literacy assessments. *Cognitive Research: Principles and Implications*, 10(1), 15.

Sokolowski, D.J., Mai, M., **Verma, A.**, Morgenshtern, G., Subasri, V., Naveed, H., Yampolsky, M., Wilson, M.D., Goldenberg, A. and Erdman, L., 2025. iModEst: disentangling-omic impacts on gene expression variation across genes and tissues. NAR Genomics and Bioinformatics, 7(1), p.lqaf011.

Verma, A., Mukherjee, K., Potts, C., Kreiss, E., and Fan, J. (2024). Evaluating human and machine understanding of data visualizations. *Proceedings of the 46th Annual Meeting of the Cognitive Science Society*.

Verma, A., Morais, L., Dragicevic, P., and Chevalier, F. (2023). Designing resource allocation tools to promote fair allocation: Do visualization and information framing matter? *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23)*.

Morgenshtern, G., **Verma, A.**, Tonekaboni, S., Greer, R., Mazwi, M., Chevalier, F., and Goldenberg, A. (2023). RiskFix: Supporting expert validation of predictive timeseries models in high-intensity settings. *EuroVis 2023-Short Papers*.

Posters and Workshops

Verma, A., Wong, L., Caren, M., Hawkins, R., Satyanarayan, A. and Fan, J.E. (2026). Unconventional communication: making meaning across modalities. *48th Annual Meeting of the Cognitive Science Society*.

Chu, J., **Verma, A.**, Davidson, G., Fraser, R., and Fan, J.E. (2025). Minds in the Making: Cognitive Science and Design Thinking. *47th Annual Meeting of the Cognitive Science Society*.

Kandapath*, M., Lee*, S., **Verma*, A.**, Chevalier, F., and Sommerville, J. (2024). Interactive causal graphics: Interaction with graphics improves learning causal associations in 5-8 years old children. *Cognitive Development Society Bi-Ennial Conference 2024*.

Gelpi, R., Ai, W., **Verma, A.**, Cunningham, W., and Sommerville, J. (2021). Dissociating children's moral evaluations and affiliative decision-making in intergroup exclusion. *Society for Research in Child Development 2021 Biennial Meeting*.

Gelpi, R., **Verma, A.**, Ai, W., Cunningham, W., and Sommerville, J. (2021). Group membership trumps moral evaluation in affiliative decision-making. *22nd Annual Meeting of the Society for Personality and Social Psychology*.

Professional and Research Experience

Research Intern *August 2023 - July 2025*

Autodesk *Toronto, ON, CA*

- Led research in Cognitive Science and Human-Computer Interaction supervised by Bon Aseniero.
- Research project studying how people plan and synthesize (mental) constraints to rearranging objects in a 3D environment.
- Built a prototype system to edit 3D scenes in-situ across different environments while mapping real time talk-aloud across different objects.

Research Lab Manager *August 2023 - July 2025*

Department of Psychology, Stanford University *Stanford, CA, USA*

- Led research in Computational Cognitive Science and Information Visualization supervised by Dr. Judith Fan.
- Built a data storage system (using TypeScript, Node.js, and MongoDB), data cleaning processes (using Pandas), evaluation tools (using Transformers, NumPy, and OpenAI), and a visualization pipeline (using Vega-Altair) to analyze responses from vision-language models. This work aimed to evaluate and improve chart understanding through question answering, resulting in a first-authored manuscript currently under review.
- Used statistical tests (using NumPy, SciPy, scikit-learn), visualizations (using Vega-Altair), and regression models (e.g., linear, logistic, mixed-effects) to compare behavioural responses and error patterns between humans and vision-language models.
- Performed analysis using latent space features, including comparing image embeddings from CLIP models and identifying common patterns in behavioral responses through Factor Analysis and Matrix Factorization methods.

Graduate Researcher *June 2022 - June 2023*

Department of Genetics and Genome Biology, The Hospital for Sick Children (SickKids)

Department of Computer Science, University of Toronto *Toronto, ON, Canada*

- Conducted research in Bioinformatics and Information Visualization supervised by Dr. Anna Goldenberg.
- Built and deployed a web-based visualization tool (using Typescript, React, D3.js, Node.js) to visualize the output from predictive time series models and record real-time model evaluation in intensive care units, leading to a publication.
- Built and deployed an interactive web-based tool (using Typescript, React, TailwindCSS, Plotly.js, Python, Flask) to visualize and summarize the contribution of different regulators to gene expression.

Graduate Researcher *Sept. 2021 - April 2023*

Dynamic Graphics Project, University of Toronto

Department of Computer Science, University of Toronto *Toronto, ON, Canada*

- Led research in Human-Computer Interaction and Information Visualization supervised by Dr. Fanny Chevalier.
- Designed and built a web-based monetary resource allocation tool (using Typescript, React, TailwindCSS, D3.js, Node.js, MongoDB) to investigate the effects of visualizations and information presentation in decision-making (using statistical contrasts and visualizations of response behaviour) leading to a first-author publication.
- Built a set of 9 games for children that communicate cause-and-effect relationships. Investigated how children interact with the game to build causal associations via interaction, leading to a poster presentation.

Research Software Developer *June 2020 - May 2023*

Department of Psychology, University of Toronto *Toronto, ON, Canada*

- Led research software development in Developmental and Cognitive Psychology supervised by Dr. Jessica Sommerville.
- Implemented a data tracking system in a Unity-based video game (using C#) to record behavioural activity.
- Developed React Native game with an embedded data tracking system to analyze user interaction data for young children.
- Developed eye-tracking software (using Typescript, React, WebGazer.js, Node.js, and ONXX) for infants and children.
- Created a platform for hosting online experiments and scheduling participants (using DigitalOcean, MongoDB, Typescript, Node.js, Tailwind CSS, and React).

Co-Founder *May 2019 - Dec. 2019*

Sparrow *Toronto, ON, Canada*

- Developed an application prototype for analyzing real-time gameplay for the League of Legends video game (using Python, PyTorch, Node.js, Prisma, AWS S3/EC2/RDS, Electron, VueJS, Apollo GraphQL, and TailwindCSS).
- Accepted in NEST program (2019 cohort) at The Entrepreneurship Hatchery Accelerator, University of Toronto. Received a second place finish in the accelerator demo day, receiving a $10k grant to aid with product development.
- Assisted in building business plan and pitch deck for potential investors and pitch competitions.

Awards & Honors

2025-26	**Presidential Fellowship**, MIT	*≈$50,000*
2019	**Hatchery Demo Day - 2nd Place Award**, University of Toronto	*$10,000*
	Hatchery Fellowship, University of Toronto	*$2,500*

Teaching

2022	**CSC318: The Design of Interactive Computational Media**, Workshop Teaching Assistant

Mentoring

2025	**Avery Yue**, Psychology, Stanford University
2024-25	**Rio Aguina-Kang**, Psychology, UCSD & Stanford University
2023	**Ofek Gross**, Computer Science, University of Toronto (Course Project, CSC494)
2022-23	**Mishaal Kandapath**, Computer Science - University of Toronto
2022-23	**Codi Lee**, Computer Science - University of Toronto
2022-23	**Emilie Ng**, Psychology - University of Toronto (Course Project, PSY299)

Service and Outreach

2026	**CogSci 2026**, Workshop Co-Organizer for *Unconventional Communication: Making Meaning Across Modalities*	*Rio de Janeiro*
2025	**CogSci 2025**, Workshop Co-Organizer for *Minds in the Making: Cognitive Science and Design Thinking*	*San Francisco*
2024	**CogSci 2024**, Workshop Co-Organizer for *COGGRAPH: Building bridges between cognitive science and computer graphics*	*Rotterdam*
2023	**ACM CHI 2023**, Student Volunteer	*Hamburg*
2022-23	**Dynamic Graphics Project Lab**, Human-Computer Interaction General Meeting Organizer	*Toronto*
2021-23	**Dynamic Graphics Project Lab**, Human-Computer Interaction Reading Group Organizer	*Toronto*
2021	**IEEE VIS 2021**, Student Volunteer	*Online*
2017	**Unverisity of Toronto**, Orientation Leader	*Toronto*
2014-15	**Kitsilano Secondary School**, Peer Tutor for 8th Grade Mathematics	*Vancouver*
2015-16	**University of British Columbia**, Geering Up Engineering Teacher Assistant	*Vancouver*